



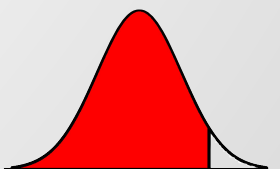
Beyond Accuracy: Stability-Oriented Learning for TCRI Credit Risk Prediction

Ying-Chieh Hsu, Kuan-Chu Wang, Jhih-Yi Chen, Huei-Wen Teng



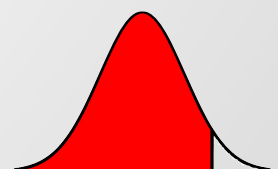
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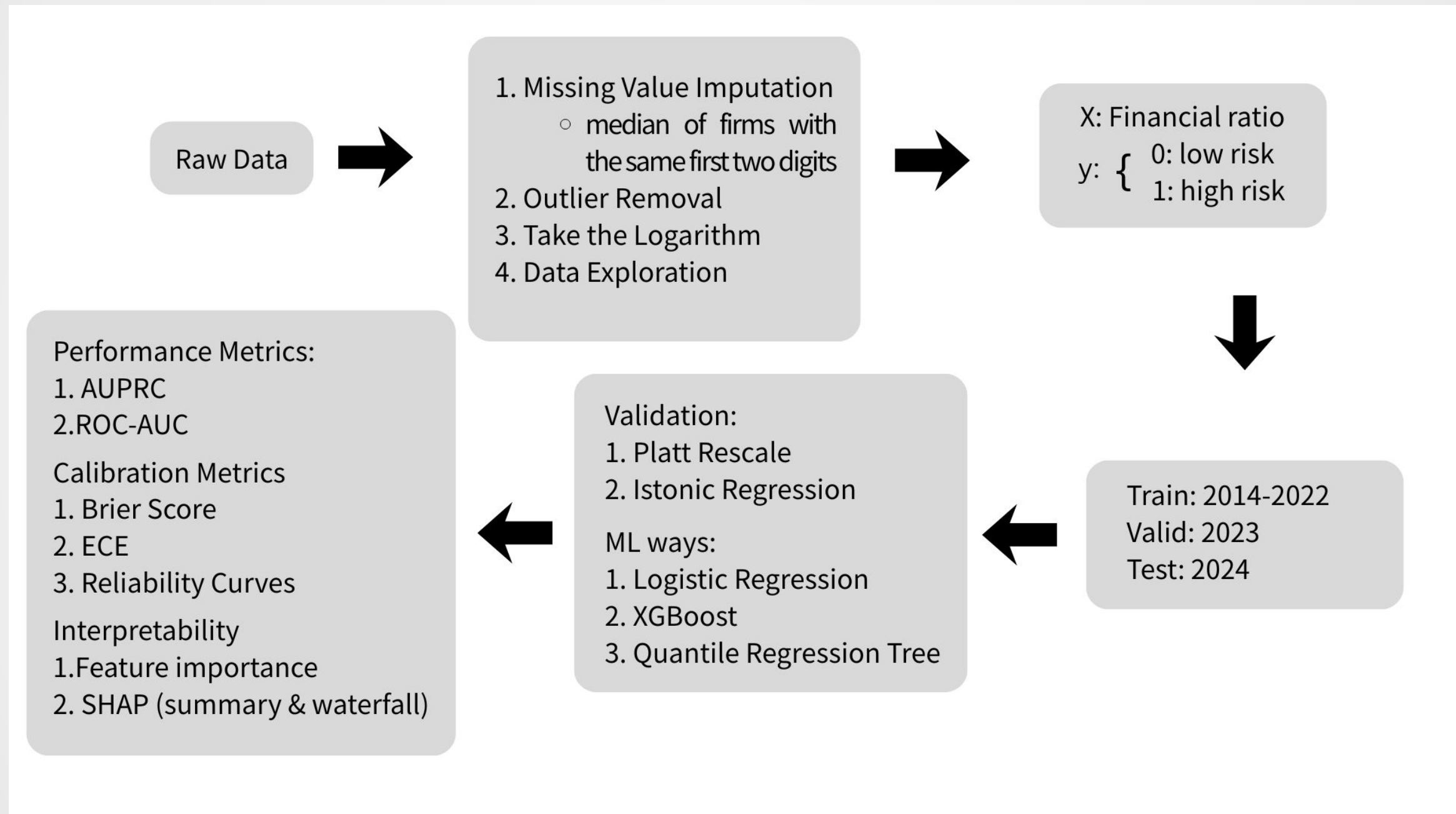


Introduction

- ▣ Credit risk prediction is crucial for financial stability
- ▣ TCRI is widely used but under-explored in academic prediction models
- ▣ Need for a transparent, interpretable, and stable credit risk framework
- ▣ Predict whether a firm will enter the high-risk category
- ▣ Reweighs industries, SHAP



Design of Research



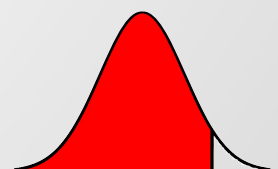
Data

- ▣ Dependent variable: TCRI score
- ▣ Independent variable: the most significant financial ratios identified in Beaver and Altman
 - ▶ Working Capital / Total Assets
 - ▶ Retained Earnings / Total Assets
 - ▶ Cash Flow / Total Debt
 - ▶ Total Debt / Total Assets
 - ▶ Current Ratio



Data Preparation and Cleaning

- ▣ To ensure data quality, we addressed missing values and removed extreme outliers before analysis.
- 1. Missing Value Imputation
 - ▶ Create a missing value indicator for each independent variables
 - ▶ imputed using the median of firms with the same first two digits of the company code.
 - ▶ If the whole group are missing → use the overall sample median
- 2. Outlier Removal
 - ▶ Observations where any ratio exceeded three standard deviations from the mean were excluded
- 3. Take the logarithm



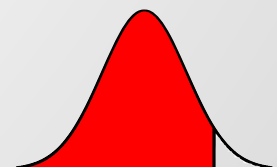
Key Descriptive Statistics

	WorkingCapital_TotalAssets	RetainedEarnings_TotalAssets	CashFlow_TotalDebt	TotalDebt_TotalAssets	CurrentRatio
NA(%)	3.13318	0	0.0964763	0	3.13318
mean	0.260084	0.111762	0.109787	0.457993	3.04299
std	0.205335	1.27573	0.49632	0.195864	11.1524
min	-0.742243	-82.0719	-18.1768	0.00303337	0.0588412
50%	0.251428	0.150736	0.080471	0.454693	1.79589
max	0.996967	1.58174	6.63415	1.05463	711.495
skewness	0.0955887	-44.8339	-10.2512	0.228632	29.3605
kurtosis	0.472559	2247.58	264.752	-0.219784	1243.48

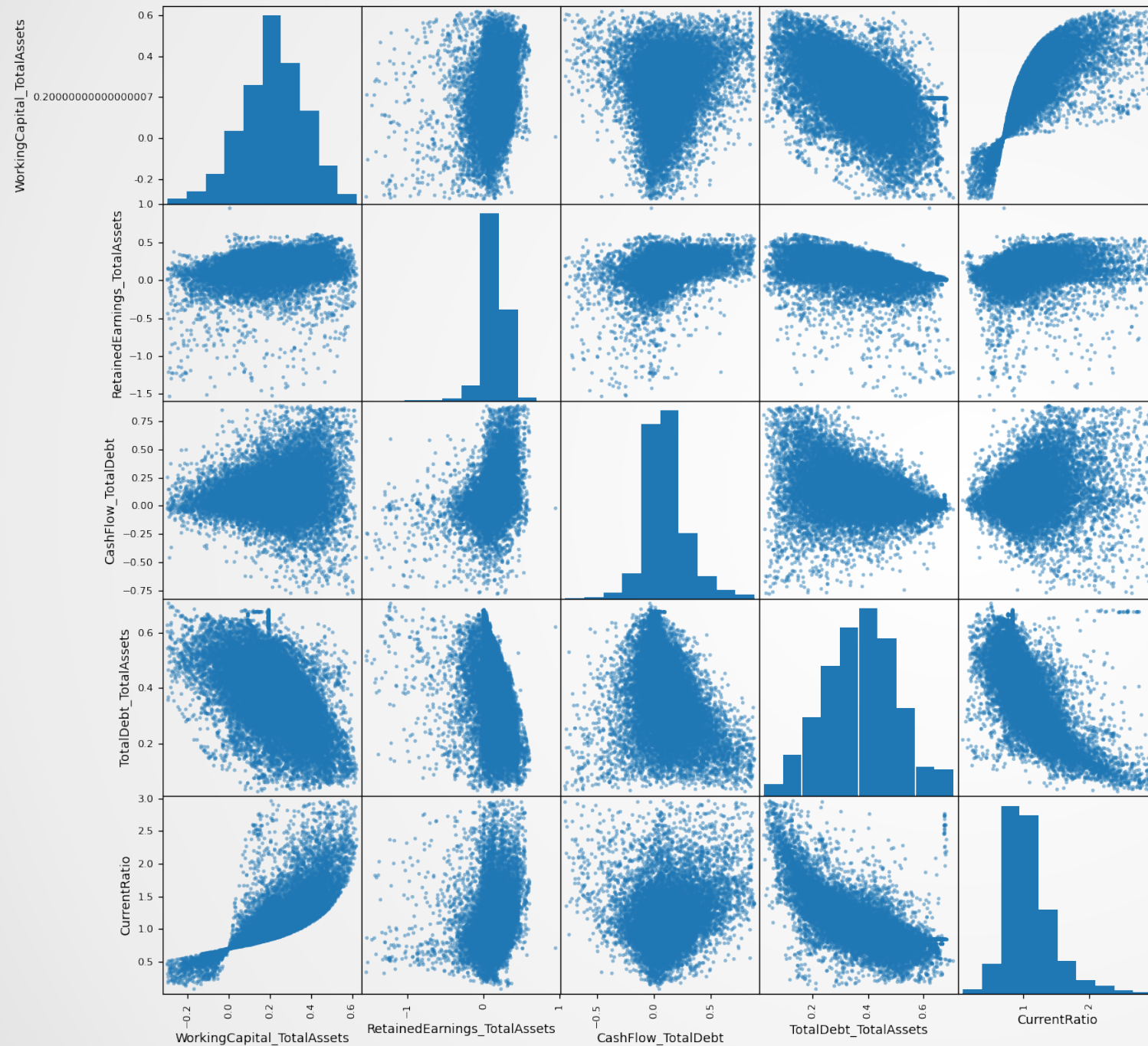


Data cleaning

	WorkingCapital_TotalAssets	RetainedEarnings_TotalAssets	CashFlow_TotalDebt	TotalDebt_TotalAssets	CurrentRatio
NA(%)	0	0	0	0	0
mean	0.215023	0.135149	0.103381	0.373306	1.08464
std	0.15288	0.167935	0.186678	0.129052	0.380631
min	-0.298188	-1.53474	-0.78052	0.0258126	0.0741276
50%	0.215281	0.141115	0.0783935	0.378056	1.00802
max	0.623079	0.948465	0.885164	0.706987	2.96362
skewness	-0.242851	-2.41152	0.543679	-0.0259596	1.34961
kurtosis	0.0358163	15.6336	2.59051	-0.345129	2.99375



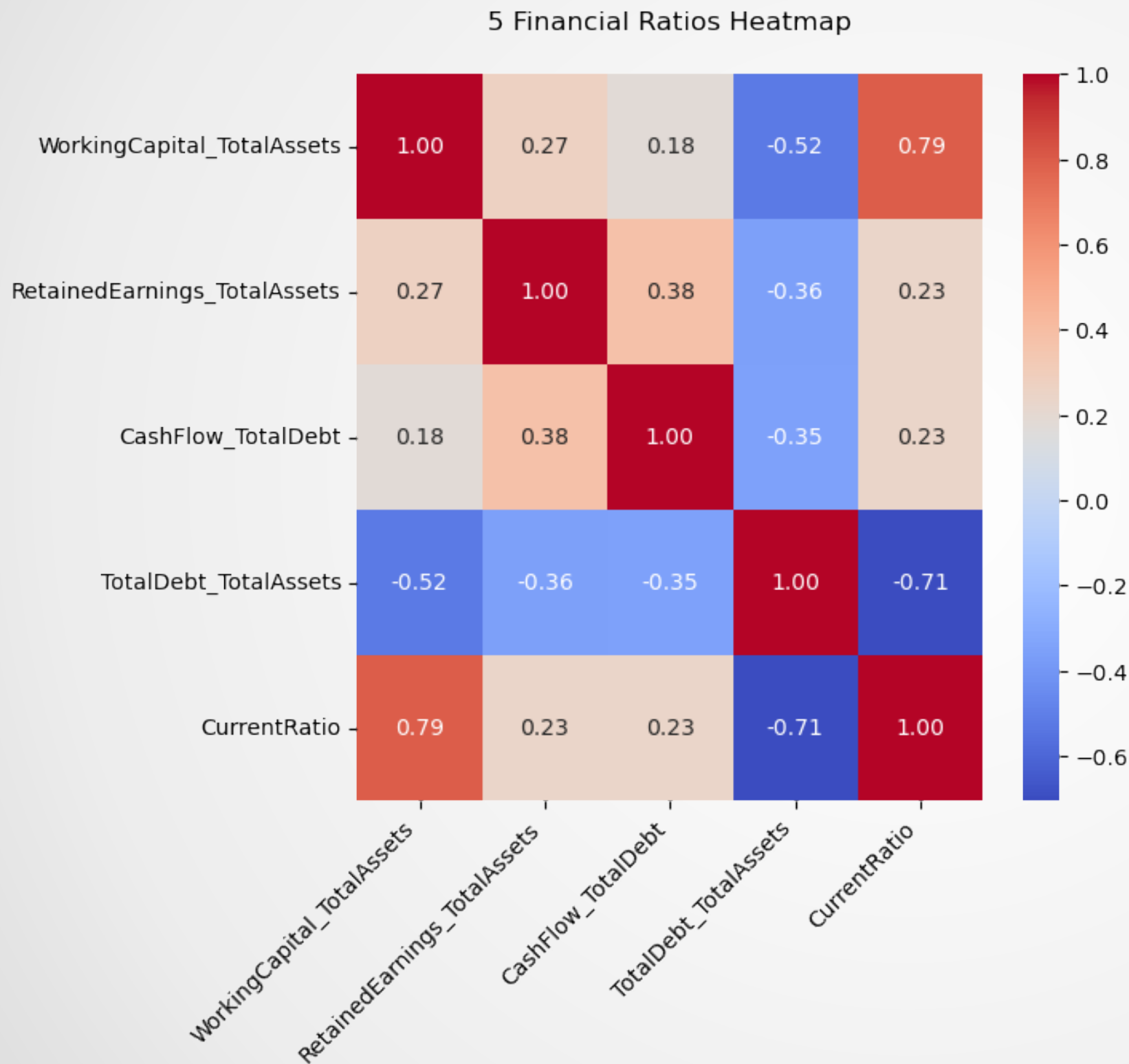
Data exploration



- Strong Linear Patterns:
 - WorkingCapital_TotalAssets and CurrentRatio (positive slope).
 - TotalDebt_TotalAssets and WorkingCapital_TotalAssets and CurrentRatio (negative slopes).
- Non-linear Relationships:
 - CurrentRatio vs WorkingCapital_TotalAssets, suggesting diminishing returns.



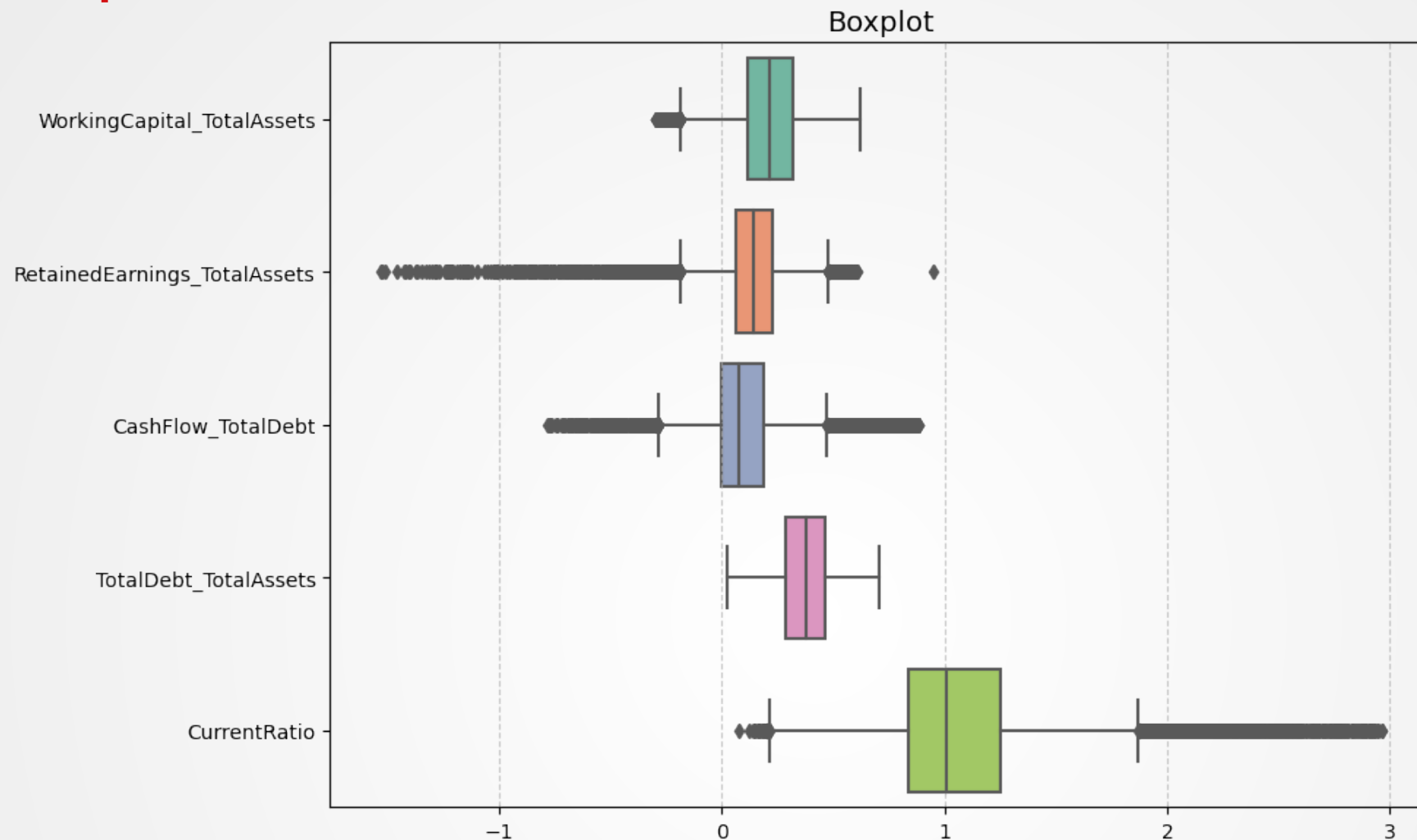
Data exploration



- High Positive Correlation:
 - WorkingCapital_TotalAssets and CurrentRatio → 0.79
 - companies with more working capital tend to have better liquidity.
- High Negative Correlation:
 - TotalDebt_TotalAssets and CurrentRatio → -0.71
 - Companies with higher debt levels typically have lower liquidity ratios.



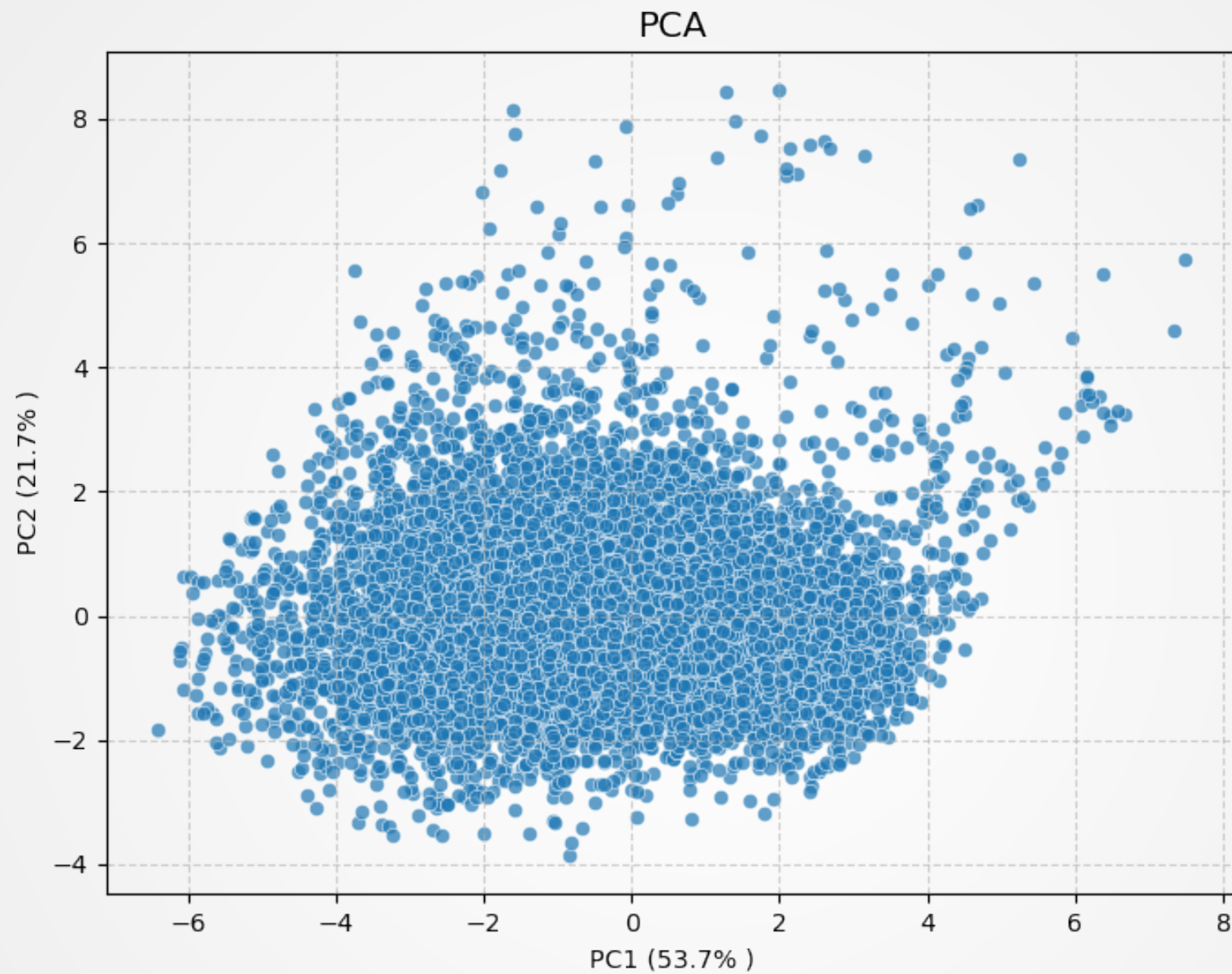
Data exploration



- Retained Earnings/Total Assets and Current Ratio show wide dispersion and many outliers → high heterogeneity in profitability and liquidity.
- Debt Ratio and Working Capital/Total Assets are more concentrated → greater consistency in capital structure.



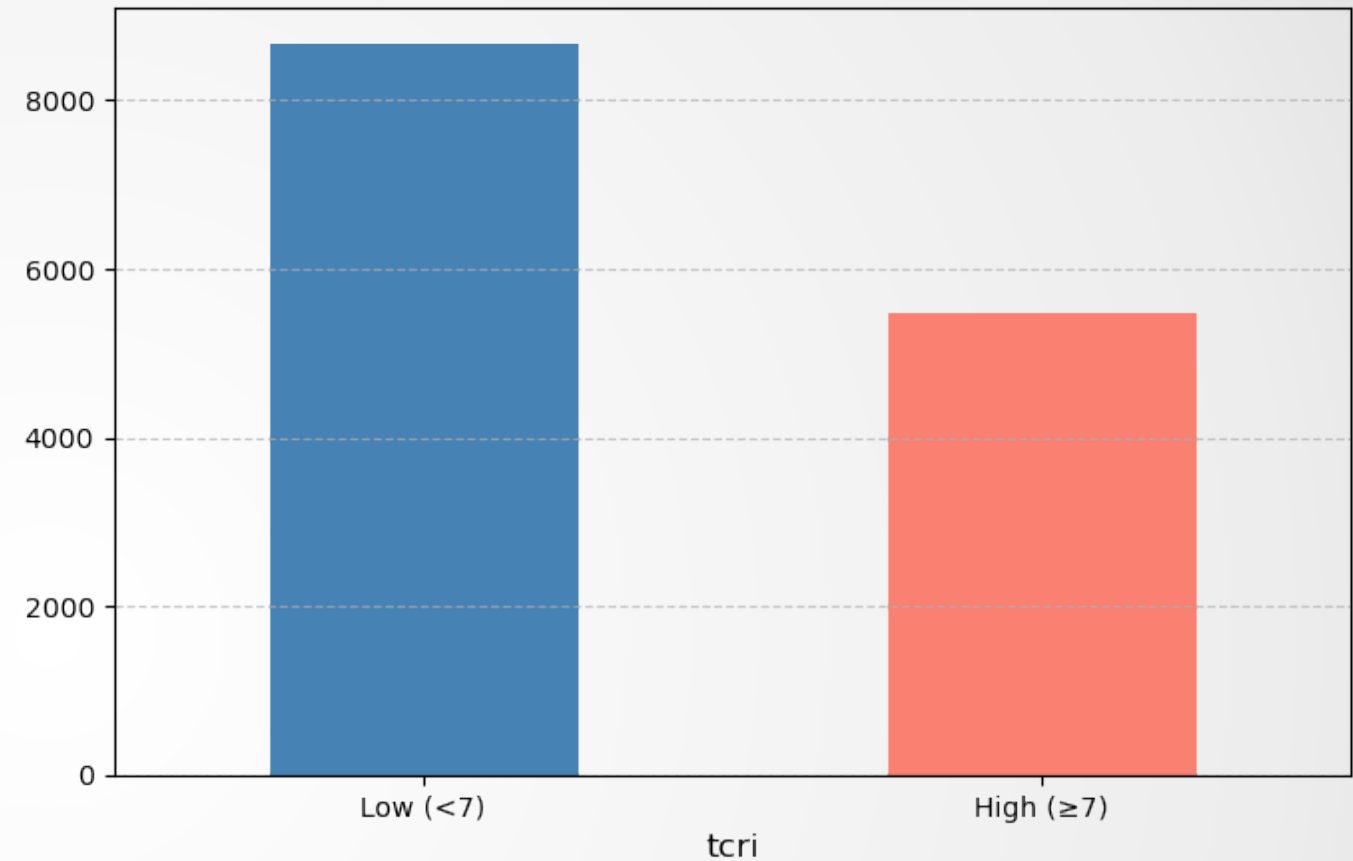
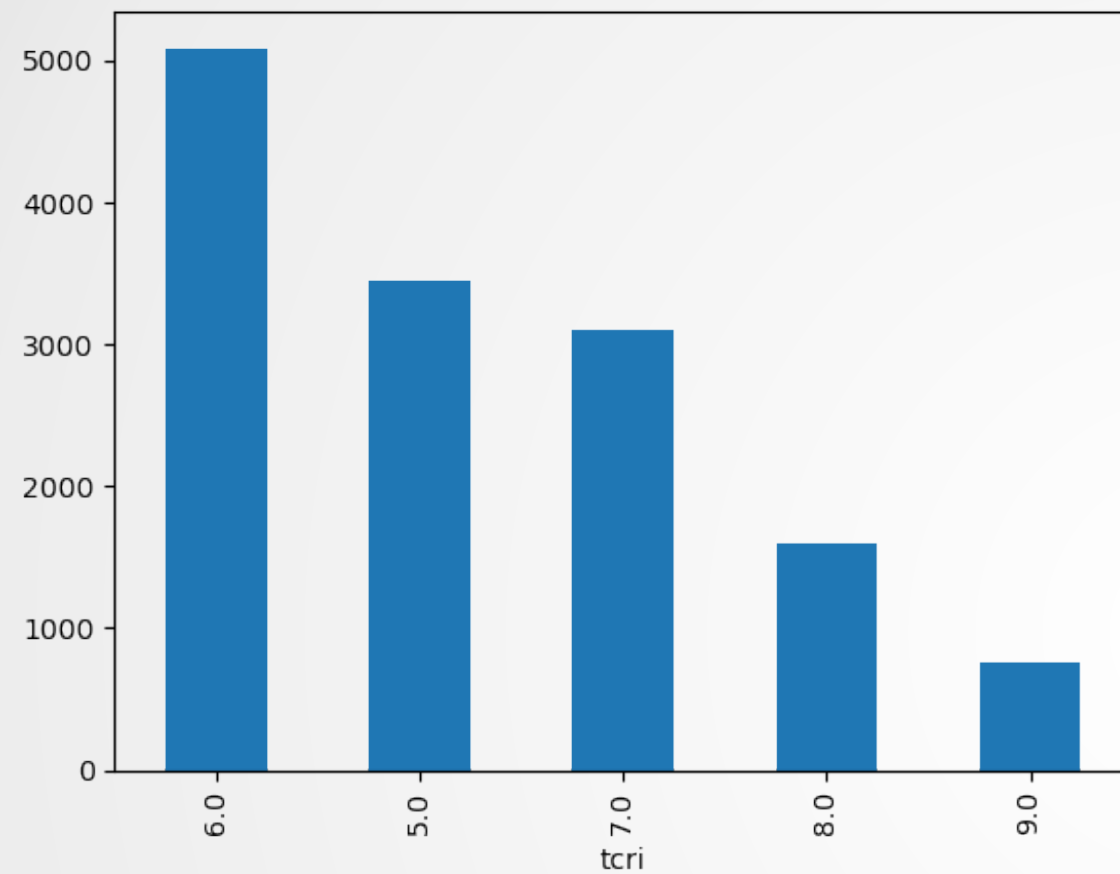
Data exploration



- No obvious clustering



Data exploration

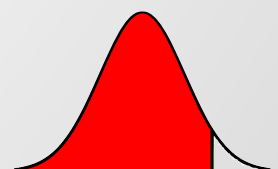


- There was no obvious uneven distribution after grouping.



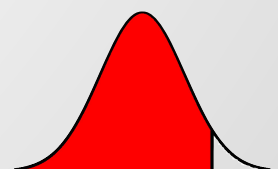
Data exploration

- ◉ Ratios provide **complementary signals**—moderate overlaps between leverage and liquidity, but largely **distinct information profiles**, supporting their joint use in **credit rating modeling** and **SHAP analysis**.



Benchmark

- ▣ Dataset (2014-2024):
 - ▣ TEJ financial ratio
 - ▣ TCRI ratings
- ▣ Use **t-year** financial ratios to predict **t+1** high-risk status (binary)
 - ▣ High risk (1): $\text{tcricri} \geq 7$
 - ▣ Low risk (0): $\text{tcricri} < 7$

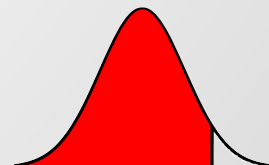


Benchmark

▣ Sample Size / Positive Rate

		n	pos	pos_rate_%
split				
2015–2022	Train	4624	1602	34.65
2023	Valid	616	221	35.88
2024	Test	619	218	35.22

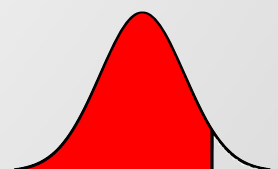
▣ Logistic Regression



Test Performance

- ▣ Platt rescale
 - ▣ Probabilities more reliable
 - ▣ Operating-point trade-off
 - ▣ Precision
 - ▣ Recall
 - ▣ F1

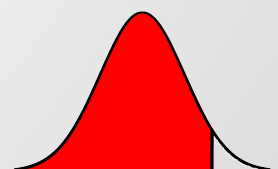
	f1	precision	recall	brier
raw	0.7017	0.6125	0.8212	0.1556
platt	0.7043	0.6788	0.7318	0.1416



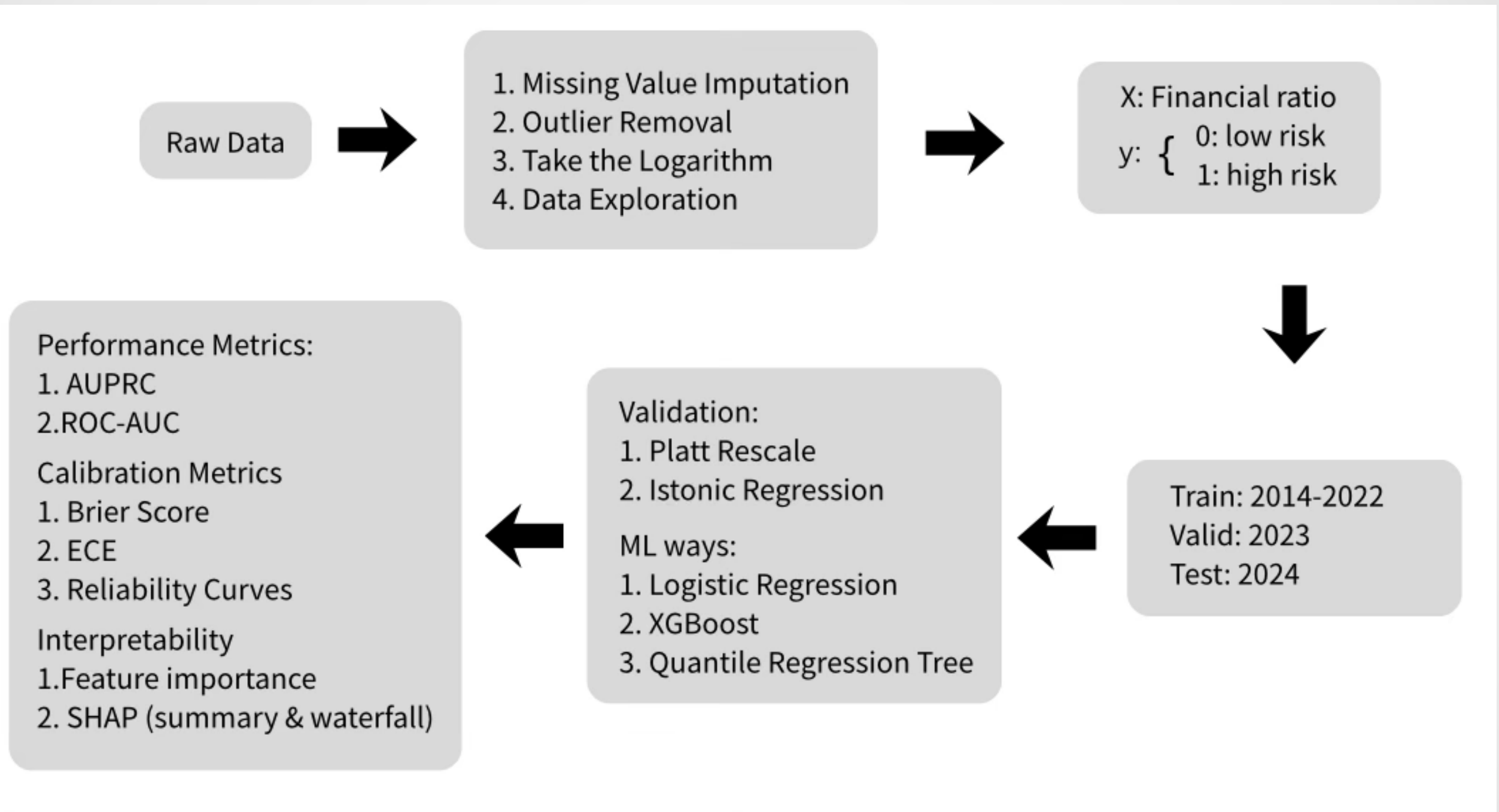
Forecast Result

- ▣ Captures **~82%** of truly high-risk
- ▣ About **60%** flagged firms are truly high-risk

	n	tp	fp	fn	tn	precision	recall	accuracy
forecast_year								
2024	619	147	93	32	347	0.6125	0.821229	0.798061



Flow chart



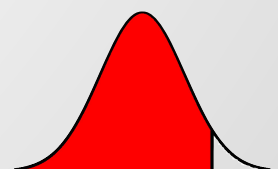
Model Training

- ▣ Logistic Regression (Benchmark model)
 - ▣ Uses L2 regularization and class weights to handle imbalance.
- ▣ XGBoost (Binary Classification)
 - ▣ Captures nonlinear relationships and feature interactions.
 - ▣ Gradient boosting trees
- ▣ Quantile Regression Trees (QRT)
 - ▣ Estimates uncertainty in predictions.
 - ▣ finds the critical quantile $\tau(x)$ where prediction switches from $0 \rightarrow 1$
 - ▣ approximates $\mathbb{P}(y=1|x) \approx 1 - \tau(x)$ to interpret prediction probability with confidence intervals.



Model Training

- ▣ Calibration
 - ▣ All models' probabilities are calibrated on a validation set.
 - ▣ Compared two methods:
 - ▣ Platt scaling
 - ▣ Isotonic regression
 - ▣ The better mapping is fixed for the test set.
 - ▣ All fitting and transformations are done only on training data to prevent leakage.



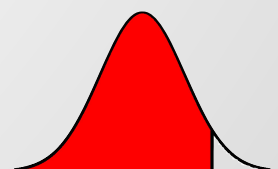
Evaluation Metrics

- ▣ Performance Metrics
 - ▣ Primary: AUPRC, ROC-AUC
 - ▣ Additional: F1, Precision, Recall, Precision@k / Recall@k
 - Evaluate model under various thresholds
- ▣ Calibration Metrics
 - ▣ Brier Score, ECE, Reliability Curves
 - ▣ Compare pre- vs post-calibration performance
- ▣ Interpretability
 - ▣ Feature importance
 - ▣ SHAP (summary & waterfall)
 - ▣ Partial dependence plots (PDP)
 - ▣ For QRT: check coverage accuracy



Reproducibility and Deployment

- Ensuring Reproducibility
 - Fix random seeds to ensure consistent results
 - Version the preprocessing pipeline and final feature set
 - Store model hyperparameters and selected calibration method (Platt or isotonic)
 - Use time-aware splits for rerunnable experiments



Reproducibility and Deployment

- ▣ Deployment / Operation Workflow
 - ▣ Each period t :
 - ▣ Ingest current data
 - ▣ Output calibrated risk scores for $t+1$
 - ▣ Generate decisions based on risk scores
 - ▣ Provide per-firm SHAP top-N contributions for explanations
 - ▣ Continuously monitor:
 - ▣ Calibration drift (Brier Score / ECE)
 - ▣ Data drift (Population Stability Index, PSI)





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